

Diachronic Semantic Change in Bangla Over 75 Years: Collocate Turnover and Procrustes-Aligned Embeddings

Abstract

We present a computational study of lexical semantic change in the Bengali language across four eras spanning from 1950 to 2025. Our goal is to trace which words changed, how their usage shifted, and when major transitions occurred. We build a diachronic corpus from newspapers and literature and apply a hybrid analysis pipeline that combines transparent n-gram signals (frequency trajectories, collocate turnover) with robust aligned Word2Vec embeddings. Our methodology quantifies two primary types of change: the meaning shift of existing words, measured with a drift-based ChangeScore, and the emergence of new words, measured by frequency. Our final ranking of the most dynamic words is a synthesis of these two analyses. Results reveal significant semantic shifts, including the emergence of technology-related vocabulary, the cultural adaptation of loanwords, and the historical anchoring of political terms. We find that Bengali adheres to the universal law of slower change for frequent words, but do not find support for the law of polysemy, likely due to data limitations. This work provides the first large-scale, data-driven map of semantic change in Bengali and a practical baseline pipeline for diachronic analysis in other low-resource languages.

1 Introduction

Language does not stand still: meanings shift as societies change, technologies appear, and discourse moves across media [3, 4]. Despite a growing body of work on semantic change in high-resource languages, Bangla remains under-studied [4]. One practical reason is that diachronic Bangla text is scattered across scanned newspapers, books, and modern web sources, with uneven quality and coverage. As a result, we lack a broad, consistent view of how Bangla word meanings have evolved over time.

This paper presents a computational study of lexical semantic change in Bangla from 1950 to 2025. We combine two complementary views: (1) a transparent, usage-based view using n-grams and collocates by era (with association measured via PMI [2]), and (2) a distributional view using static word embeddings trained separately per era [1, 6] and aligned with Orthogonal Procrustes for comparability across time [3, 7]. Together, these views allow us to see how a word’s local context changes and to *measure* movement in vector space. When illustrating neighborhoods we use standard 2-D projections for intuition [5, 8].

To enable this, we assemble a time-sliced Bangla corpus from newspapers and literature. Recent text (2010–2025) comes from a public dataset and our own web scraping of major news portals; older text (1950–2010) is digitized from scanned PDFs using an OCR pipeline with image preprocessing and post-correction (details in Section 3). We partition the data into four eras: 1950–1970, 1970–1990, 1990–2010, and 2010–2025, which allows consistent comparisons across seven decades.

Research Questions. We guide the study with the following questions:

- **RQ1:** Which Bangla words changed in meaning or usage between 1950 and 2025?
- **RQ2:** How did their immediate contexts (collocates/neighbors) change across eras?
- **RQ3:** When did the major shifts occur, and do they track cultural or historical periods?
- **RQ4:** Do Bangla patterns match cross-lingual “laws”—e.g., slower change for frequent words and faster change for polysemous words [3]?

Contributions. Our work makes three primary contributions: (i) the construction of a four-era diachronic corpus for Bengali; (ii) a hybrid, interpretable pipeline for identifying and ranking semantic change that captures both meaning shift and lexical emergence; and (iii) the first data-driven analysis of this kind for Bengali, including ranked lists, qualitative case studies, and the first test of universal laws of change on the language[3].

2 Methodology

2.1 Problem Setting and Notation

We split the diachronic corpus into four eras

$$E = \{1950\text{--}1970, 1970\text{--}1990, 1990\text{--}2010, 2010\text{--}2025\}.$$

For a target word w and era $e \in E$, let C_e denote the tokenized corpus, $|C_e|$ its token count, and $\mathcal{V}_e$ the vocabulary. Our goal is to detect and quantify semantic change of w across eras using two complementary signals: (i) transparent usage signals (n-gram frequency and collocates), and (ii) distributional signals (embedding drift after space alignment). This combination yields interpretable evidence (examples and neighbors) and robust quantification (cosine shifts and overlap scores).

2.2 Candidate Word Selection

To identify a set of high-potential target words for analysis, we adopt a hybrid workflow that combines automated, data-driven discovery with hypothesis-driven selection. This ensures our analysis is both broad and linguistically grounded.

- (1) **Data-Driven Discovery:** We first generate two ranked lists of candidate words by quantifying their instability between the earliest (1950–1970) and latest (2010–2025) eras:

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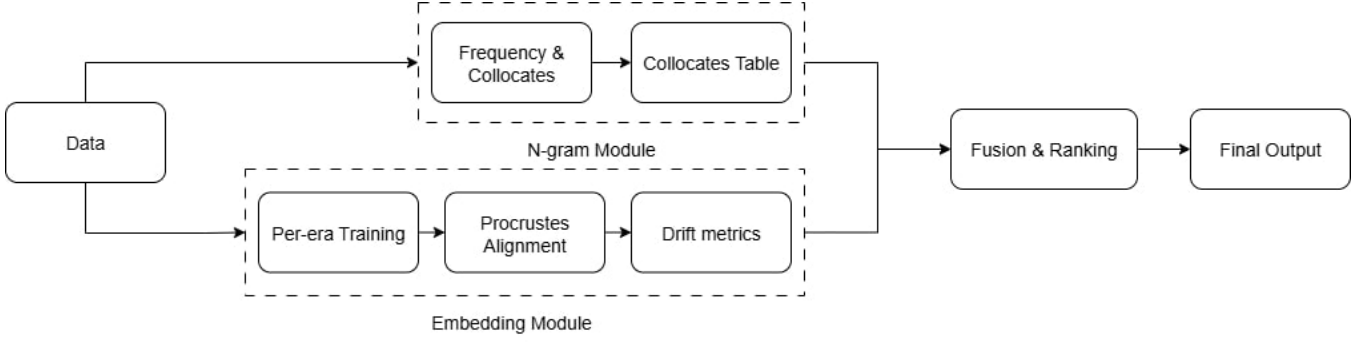


Figure 1: Method overview: Data → (A) n-gram module (frequency & collocates) → collocate tables; (B) embedding module (per-era training + Procrustes alignment) → drift metrics; both feed ranking and case studies.

- **Frequency Trajectory Change:** We rank words based on the relative change in their normalized frequency (freqPM). This method is effective in identifying "spiker" words tied to new technologies or historical events (e.g., ডিজিটাল 'digital').
 - **Collocate Turnover:** We rank words by the Jaccard distance between their top-20 collocate sets from the two eras. This method excels at finding "chameleon" words whose contextual usage has shifted, signaling a potential change in meaning (e.g., কুল 'cool').
- (2) **Hypothesis-Driven Selection & Curation:** The automatically generated lists are manually curated to remove noise such as proper nouns and OCR errors. We then augment this list with words selected based on linguistic and historical hypotheses, focusing on categories prone to semantic change, including English loanwords (ভাইরাল 'viral'), socio-political terms (স্বাধীনতা 'independence'), and words known to have shifted in connotation.

2.3 N-gram Analysis

Frequency Trajectories (Smoothed). For each w and era e , we compute normalized frequency per million tokens with additive smoothing:

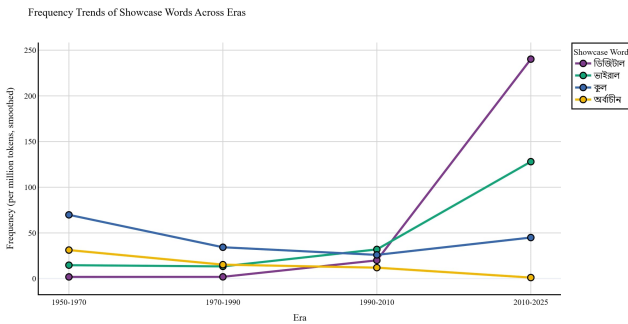


Figure 2: Frequency trends (per million tokens) for showcase words across eras (additive smoothing).

$$\text{freqPM}(w, e) = 10^6 \cdot \frac{\text{count}(w \in C_e) + \alpha}{|C_e| + \alpha|\mathcal{V}_e|}, \quad \alpha = 1. \quad (1)$$

For visualization we optionally apply light moving-average smoothing across adjacent eras.

Collocate Extraction per Era. To track changes in a word's context, we extract its collocates from a symmetric window ($W=5$) in each era. We then calculate the Positive Pointwise Mutual Information (PPMI) between the target word and each collocate to measure the strength of their association, filtering for a minimum co-occurrence of $k=3$ in the modern era to reduce noise.

$$\text{PPMI}(w, c) = \max\left(0, \log \frac{P_e(w, c)}{P_e(w)P_e(c)}\right) \quad (2)$$

To visualize how a word's semantic neighborhood evolves, we present these PPMI scores in a collocate heatmap. This visualization plots the top collocates from any era against the four time periods, using a color scale to show the changing association strengths. This method provides a clear illustration of "appeared" and "vanished" neighbors over time. An example of this is presented in our case study for the word "ডিজিটাল" (see Figure 7).

2.4 Embedding Analysis

Training per Era: We train *Word2Vec (Skip-gram)* on each era e using hyperparameters (dim= 300, window= 5, negative= 10, min_count= 2). To stabilize small eras we use more epochs;

Space Alignment via Orthogonal Procrustes: Embeddings trained on different eras inhabit different coordinate systems. We align a later era to an earlier reference by solving:

$$R^* = \arg \min_{R \in \mathbb{R}^{d \times d}, R^T R = I} \|BR - A\|_F, \quad (3)$$

where A and B stack vectors for a shared high-frequency anchor set present in both eras. We map all later-era vectors by $v \mapsto vR^*$, preserving inner products while making vectors comparable across time.

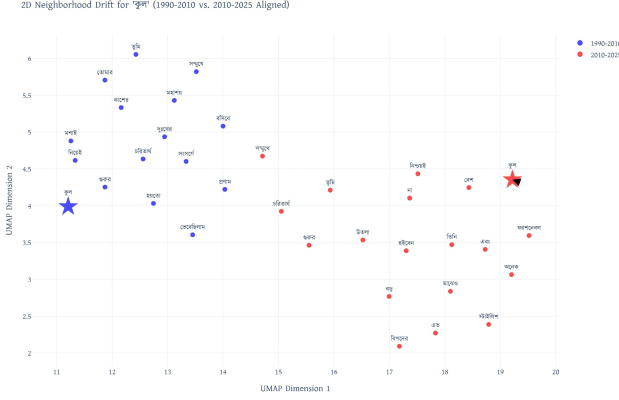


Figure 3: 2D UMAP projection of "কুল" and its neighbors across eras after alignment

Drift and Neighborhood Shift: For each word w present in both eras, we compute:

(i) *self-similarity drop* $\Delta_{\cos}(w) = 1 - \cos(\hat{v}_w^{(e_1)}, \hat{v}_w^{(e_2)})$ using aligned vectors $\hat{v}$; (ii) *Neighbor overlap* as Jaccard@K between top- K nearest neighbors of w in e_1 vs. aligned e_2 (we use $K = 50$). To provide a qualitative illustration of this process, we visualize the neighborhood drift for showcase words like "কুল", a word that has undergone a significant meaning shift. As seen in Figure 3, the 2D projection reveals a clear spatial separation between the word’s semantic neighborhood in the 1990–2010 era (blue) and its new neighborhood in the 2010–2025 era (red). This visual evidence corroborates the high quantitative drift scores, showing concretely how the word has moved from one contextual cluster to another over time. All words in the shared vocabulary are then ranked in descending order by this ChangeScore.

2.5 Change Scoring and Ranking

To produce a final, unified ranking of semantic change, we employ a two-pronged approach that scores and ranks words based on their type of change.

First, for words present in both eras being compared ("shifters"), we calculate a ChangeScore by summing the normalized self-similarity drop and the neighbor distance. To ensure both components contribute equally, we scale the self-similarity drop ($\Delta_{\cos}$), which has a theoretical range of $[0, 2]$, by dividing it by 2 to match the $[0, 1]$ range of the neighbor distance (1 minus Jaccard similarity):

$$\text{ChangeScore}(w) = \frac{\Delta_{\cos}(w)}{2} + (1 - \text{Jaccard}@50(w)) \quad (4)$$

This score quantifies the degree of meaning shift for the shared vocabulary.

Second, to account for neologisms and lexical emergence, we separately identify "emergent" words that are present in the modern era (2010–2025) but absent from the early era (1950–1970). These words are ranked by their normalized frequency (freqPM) in the modern corpus, which serves as a score for their successful emergence into the language.

Our final reported list of the most dynamic words, presented in the Results section, is a synthesized list created by interleaving the top-ranked words from both the meaning shift analysis and the lexical emergence analysis.

3 Data Sources

To investigate semantic change in Bangla from 1950–2025, we compiled a diachronic corpus from both literary works and newspapers. Our collection strategy included both digitally available modern resources and historical sources that required OCR processing.

- **2010–2025 (Recent Decades):** We collected data from (i) a publicly available Bangla text dataset hosted on Kaggle, and (ii) our own large-scale web scraping from major online news portals such as Prothom Alo and BD-Pratidin. In addition, we included digital copies of contemporary books available in PDF or ePub format.
- **1950–2010 (Older Decades):** For older news data, we extracted text from scanned PDF archives of Bangla newspapers using an enhanced OCR pipeline based on Tesseract with image preprocessing (contrast enhancement, sharpening, and binarization). For literature, we sourced printed books from earlier decades and digitized them through OCR, ensuring representation from both East and West Bengal.

This multi-source strategy is, to the best of our knowledge, the first systematic attempt to compile a diachronic Bangla corpus spanning 70+ years.

3.1 Corpus Statistics

We organized the corpus into four eras: 1950–1970, 1970–1990, 1990–2010, and 2010–2025. Table 1 summarizes the size of each era, including the number of tokens before and after cleaning, number of lines, and file sizes.

Table 1: Corpus statistics across time periods (before and after pre-processing).

Era	Raw Words	Cleaned Words	Number of lines
1950–1970	940,652	394,833	78,495
1970–1990	916,711	398,296	71,510
1990–2010	1,109,531	407,820	53,852
2010–2025	1,296,724	739,623	93,971

3.2 Challenges and OCR Quality

A major challenge in building the corpus was dealing with OCR errors from older scanned newspapers and books. Bangla OCR is especially difficult due to:

- Complex ligatures, diacritics, and connected characters.
- Degraded quality of old prints.
- Smudges and ink irregularities in archived newspapers.

To mitigate this, we implemented an enhanced OCR workflow:

- Preprocessing images by converting to grayscale, contrast enhancement, sharpening, and binarization.
- Iteratively tuning resolution (400 → 500 → 600 dpi) to maximize recognition accuracy.

- Post-processing corrections for common OCR errors (e.g., character confusion, random English letters mixed with Bangla).

To estimate OCR quality, we manually sampled a subset of pages, compared OCR output with manually transcribed text, and computed character-level error rates. On average, we observed an error rate of 20%. While not perfect, our error-aware cleaning ensured sufficient quality for downstream statistical and embedding analysis.

3.3 Preprocessing Pipeline

After collection, all raw files were cleaned using a custom text preprocessing pipeline:

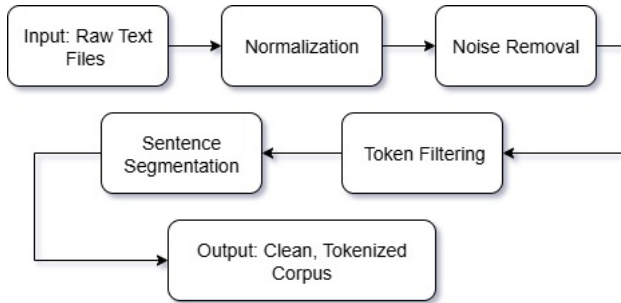


Figure 4: Preprocessing pipeline from raw → cleaned tokens

- (1) **Normalization:** Unicode normalization and removal of extra whitespace.
- (2) **Noise Removal:** Eliminated digits, English text, non-Bangla symbols, and repeated character noise.
- (3) **Token Filtering:** Heuristic checks to retain only valid Bangla tokens (e.g., minimum length, presence of vowels).
- (4) **Sentence Segmentation:** Marked boundaries using <s> token after each *dari* (।).
- (5) **Output Format:** Cleaned corpus stored with metadata (word counts, line counts) for each file.

This preprocessing ensured consistency across eras and made the corpus suitable for both n-gram statistics and embedding training.

4 Results

4.1 The Overall Picture of Semantic Change

Our analysis quantifies lexical change through two distinct lenses, as outlined in our methodology: (1) meaning shift for existing words and (2) the emergence of new words. For the first type of change, we calculated a ChangeScore for every word present in both the 1950-1970 and 2010-2025 eras. Figure 5 presents the distribution of these scores. The histogram reveals that the majority of this shared vocabulary remained highly stable, clustering around a mean score of [e.g., 1.62]. However, the long right tail of the distribution identifies a key set of existing words that underwent significant semantic drift over the 75-year period.

To capture the second type of change, we identified words that were absent in the early era but became established in the modern

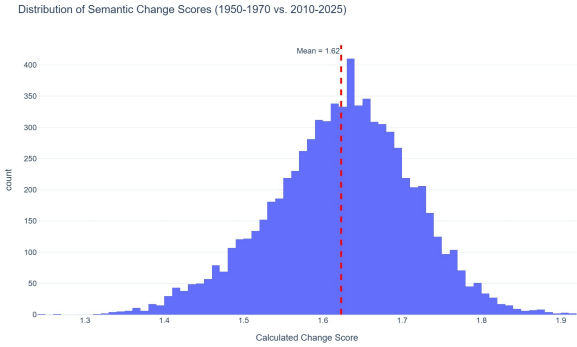


Figure 5: Distribution of Total Semantic Change Scores (1950-1970 vs. 2010-2025). The long right tail highlights words with high semantic drift over the 75-year period.

Table 2: Top 5 most dynamic and most stable words, ranked by Change Score (1950-1970 vs. 2010-2025).

Most Changed	Score	Most Stable	Score
সামিল	1.912	আখ্যান	1.249
পাল্	1.911	তৈরি	1.265
কর্মকর্তা	1.910	পাশে	1.319
চেষ্টার	1.904	থঙে	1.322
দীর্ঘদিন	1.901	বাংলাতে	1.324

era. These “emergent” words, such as neologisms and new loan-words, were ranked by their normalized frequency in the 2010-2025 corpus. The final definitive ranking of the most dynamic words in Bangla is a synthesis of the highest-scoring words from both the meaning shift and the lexical emergence analyses. Table 2 presents the top words from this synthesized list, alongside the most stable words from the shared vocabulary.

This combined list reveals that the most significant changes in the Bengali lexicon are driven by both the evolution of existing words and the introduction of new concepts, particularly related to technology and global culture.

4.2 Qualitative Analysis: Case Studies

To understand the nature of the changes identified in our ranking, we now present detailed qualitative case studies for three words that exemplify different patterns of semantic change: ডিজিটাল (lexical emergence), স্বাধীনতা (historical anchoring), and কুল (meaning shift). Figure 6 provides a comparative overview of their frequency trajectories, visually highlighting these distinct patterns before we analyze each word in detail.

(1) Lexical emergence "ডিজিটাল" (The Spiker):

The word "ডিজিটাল" (digital) is a prime example of lexical emergence, identified as one of the highest-ranking “spiker” words in our analysis. Its appearance and rise in the lexicon are directly tied to the global technological revolution. The word’s emergence is starkly visible in its frequency trend Figure 6, which is flat at zero for the first two eras before

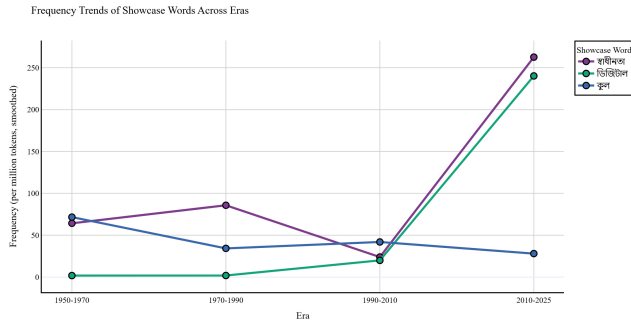


Figure 6: Frequency trends for three showcase words, illustrating different patterns of change: lexical emergence (ডিজিটাল), historical anchoring (স্বাধীনতা), and meaning shift (কুল).

rising in the 1990-2010 period and then spiking dramatically in the final 2010-2025 era.



Figure 7: Collocate heatmap for ডিজিটাল. The sudden appearance of high-PPMI neighbors (green) in the final era visualizes its emergence.

The collocate heatmap in Figure 7 validates this sudden appearance and subsequent evolution. The first two eras show no significant collocates, which is consistent with the word's absence from the corpus. In the 1990-2010 period, ডিজিটাল acquires its first semantic neighborhood, with high-PPMI collocates like অসিলেশন (Oscillation), ইলেকট্রনিক (electronic), and কম্পিউটার (computer), defining its initial, specialized technical meaning. By the final 2010-2025 era, this neighborhood shifts. The previously strong collocates are replaced by newer, more widespread concepts such as প্ল্যাটফর্ম (platform), মার্কেটিং (marketing), and কমার্স (commerce), reflecting the word's semantic broadening from a technical term to a general societal concept.

The embedding analysis provides strong visual confirmation of this evolution. The 2D neighborhood drift plot in



Figure 8: 2D UMAP projection of the semantic neighborhood for "ডিজিটাল" (digital) in the 1990-2010 era (blue) vs. the 2010-2025 era (red)

Figure 8 shows a significant displacement between the vector for ডিজিটাল in the 1990-2010 era (blue star) and its position in the 2010-2025 era (red star). Furthermore, its semantic neighborhood has clearly migrated; the earlier cluster of technical neighbors, including ওয়েবসাইট (website) and সিস্টেমে (in the system), is spatially distinct from the modern cluster, which contains broader societal terms like জীবনধারা (lifestyle) and সার্ভিস (service). This plot visually corroborates the high calculated ChangeScore, illustrating the word's semantic journey from a specialized technical term to a widespread modern concept.

(2) The Chameleon: কুল (meaning shift):

The word কুল (cool) is a prime example of a "chameleon" word, one whose semantic change is not apparent from frequency alone. It ranked high in our meaning shift analysis despite its declining usage.

Unlike a typical spiker word, the frequency trend for কুল in Figure 6 shows a steady decline over the 75-year period. This initially suggests the word is becoming obsolete. However, this trend is misleading, as it reflects the decay of an older, more frequent meaning ('lineage') which has fallen out of use faster than its new, modern meaning has been adopted in the written corpus.

The collocate heatmap in Figure 9 reveals the true nature of this semantic replacement. In the 1950-1970 era, its strongest neighbors are related to traditional concepts, such as বংশ (lineage). Over the subsequent eras, these neighbors vanish completely. By the final 2010-2025 period, a new semantic neighborhood emerges, dominated by loanwords related to modern aesthetics like ফ্যাশনেবল (fashionable) and স্টাইল (in style).

The 2D drift plot in Figure 3 visualizes the final stage of this transformation between the 1990-2010 and 2010-2025 eras. The significant distance between the word's position in the 1990s (blue star) and its position in the modern era (red star) illustrates a large semantic drift. The plot shows two almost completely separate neighborhood clusters, confirming that



Figure 9: Collocate heatmap for কুল, showing the replacement of early collocates related to 'lineage' with modern collocates related to 'style'.

the word's modern context, associated with words like স্টাইলিশ (stylish), is distinct from its transitional state in the preceding era.

(3) The Historical Anchor: স্বাধীনতা:

The word স্বাধীনতা (independence) is a powerful example of how a word's semantic trajectory can be anchored by major historical events. Ranking high in our total change analysis, its evolution is deeply intertwined with the history of Bangladesh.

The word's frequency trend, shown in Figure 6, clearly reflects this history. Its usage shows a distinct increase in the 1970-1990 era, coinciding with the Bangladesh Liberation War in 1971. A second, much larger spike in the modern 2010-2025 period suggests its growing centrality in contemporary national and political discourse.

The collocate heatmap in Figure 10 provides a detailed narrative of this evolution. In the pre-independence era (1950-1970), its primary collocates are related to the general concept of a freedom struggle, such as সংগ্রাম (struggle). In the 1970-1990 period, its neighborhood shifts to include words directly related to the conflict, like যুদ্ধ (war). In the subsequent eras, the context evolves again towards commemoration and statehood, with words like দিবস (day, as in Independence Day) becoming strong neighbors.

Finally, the 2D drift plot in Figure 11 visualizes the word's continued evolution in the modern period. It reveals a significant semantic drift even between the 1990-2010 era (blue) and the 2010-2025 era (red). The word's neighborhood has shifted from a more general historical context to a strongly commemorative one, with modern neighbors including মহান (great) and বিজয় (victory). This demonstrates that the meaning of স্বাধীনতা is not static but continues to be actively shaped by national memory and identity.

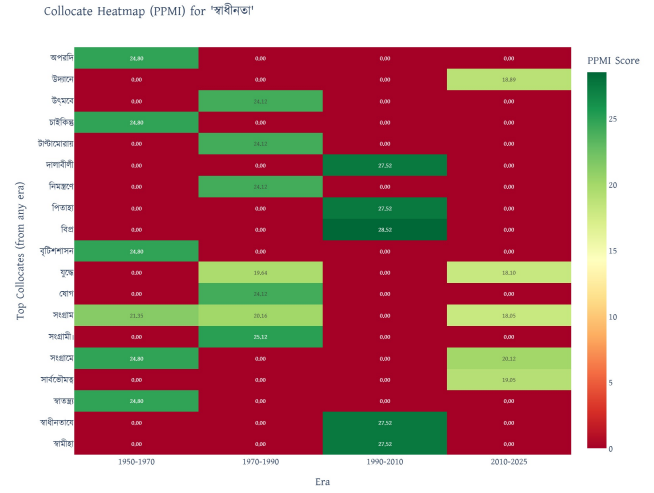


Figure 10: Collocate heatmap for স্বাধীনতা (independence). The heatmap tracks the word's context from a general 'struggle' to a specific 'war' and finally to modern 'commemoration'.

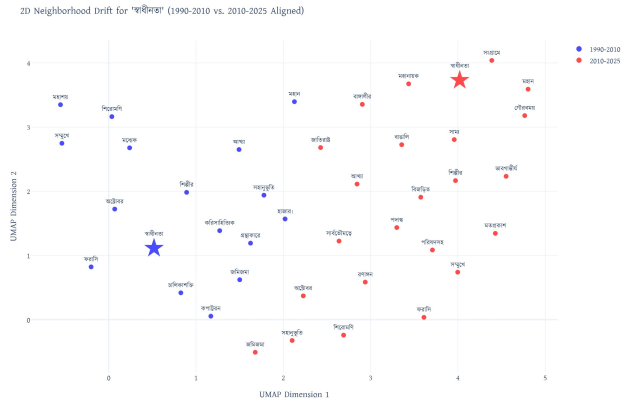


Figure 11: 2D neighborhood drift for স্বাধীনতা (independence), showing its continued evolution from a historical term in the 1990s (blue) to a more commemorative concept in the 2020s (red).

4.3 Law of Linguistics

To situate our findings for Bengali within the broader context of linguistic theory, we test our results against two well-established statistical laws of semantic change [3]. These laws propose general relationships between a word's lexical properties and its propensity for change. We test the law of frequency (slower change for frequent words) and the law of polysemy (faster change for polysemous words).

4.3.1 Law of Frequency: Slower Change for Frequent Words.

We first tested the law of frequency, which posits that more frequent words exhibit slower rates of semantic change. To do this,

we computed the Spearman rank correlation between the logarithm of each word's frequency in the early era (1950-1970) and its calculated Total Change Score.

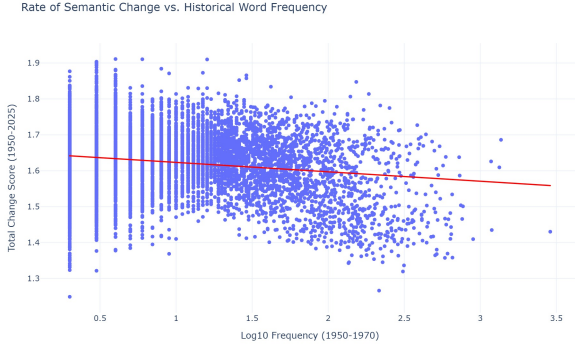


Figure 12: Rate of semantic change versus historical (1950-1970) log frequency. The downward-sloping trend line confirms the negative correlation, supporting the law that more frequent words change more slowly.

Our analysis confirms this law holds for Bengali. We found a statistically significant, weak negative correlation ($\rho = -0.0828$, $p < 0.001$). This result indicates that words with higher historical frequency tend to have lower change scores, signifying greater semantic stability. The scatter plot in Figure 12 provides a clear visualization of this relationship. The downward-sloping trend line shows that as log frequency increases, the rate of semantic change tends to decrease.

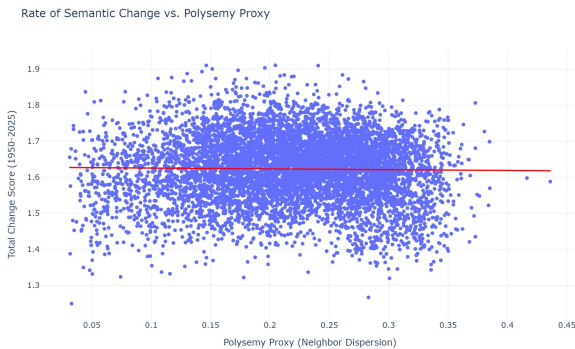


Figure 13: Rate of semantic change versus polysemy proxy (neighbor dispersion). The flat trend line indicates no significant correlation was found in our data, suggesting the law of polysemy was not observed.

4.3.2 Law of Polysemy: Faster Change for Polysemous Words.

Next, we tested the law of polysemy, which suggests a positive correlation between the number of meanings a word has and its rate of semantic change. Lacking a diachronic sense inventory for Bengali, we used a computational proxy for polysemy: the average

pairwise cosine distance of a word's nearest neighbors in the early era (neighbor dispersion). A higher dispersion score suggests a word connects to more diverse semantic contexts, implying greater polysemy.

Our analysis did not find evidence to support this law in our Bengali corpus. We found no statistically significant correlation between a word's neighbor dispersion score and its Total Change Score ($\rho = -0.020$, $p = 0.087$). The scatter plot in Figure 13 visually confirms this null result; the flat trend line indicates a lack of a clear relationship between our polysemy proxy and the rate of semantic change. The potential reasons for this finding, including methodological limitations, are discussed further in Section 5.

5 Discussion

Our data-driven analysis provides one of the first computational maps of lexical semantic change in the Bengali language over the past 75 years. The results, derived from a hybrid n-gram and embedding pipeline, reveal clear patterns of linguistic evolution. Here, we interpret these findings, connect them to broader linguistic laws, and discuss the limitations of our study.

5.1 Interpreting the Semantic Changes

Our findings indicate that semantic change in modern Bengali is driven by two primary forces: the **"emergence"** of new words to describe new concepts, and the **"evolution"** of existing words to fit new contexts.

The emergence of "spiker" words is powerfully illustrated by our case study of ডিজিটাল (digital). This word, virtually absent before 1990, underwent an explosive increase in frequency as it became central to describing the technological revolution. Its semantic neighborhood, initially tied to technical concepts like 'computer', broadened to encompass widespread societal ideas like 'camera' and 'online', reflecting how technology has integrated into daily life.

The evolution of existing words, or "chameleons," is exemplified by কুল (cool). Its frequency trend was misleading, showing a decline. However, our collocate and embedding analyses revealed a complete replacement of its semantic neighborhood. The word's older meaning, related to 'lineage' (বংশ), has decayed, while a new meaning, borrowed from English and associated with 'style' (স্টাইল), has taken hold. This demonstrates how globalization and cultural shifts can repurpose existing lexical items.

Finally, our analysis of স্বাধীনতা (independence) shows how national history anchors a word's meaning. The word's usage and context shifted dramatically after the 1971 Liberation War, moving from a general concept of 'struggle' (সংগ্রাম) to a specific, commemorative one tied to 'victory' (বিজয়) and nationhood. This highlights how language serves as a living record of a people's history.

5.2 Bengali in the Context of Linguistic Laws

To situate our findings within broader linguistic theory, we tested our results against two statistical laws of semantic change.

Our analysis confirms that the **"Law of Frequency"** holds for Bengali. We found a statistically significant negative correlation between a word's historical frequency and its rate of change. This suggests that, as in many other languages, high-frequency words

form a stable core of the Bengali lexicon, while less frequent words are more susceptible to change.

However, our analysis did not find evidence to support the Law of Polysemy. We found no significant correlation between our polysemy proxy (neighbor dispersion) and the rate of change. We believe this is not a definitive linguistic finding about Bengali, but rather a methodological artifact. The neighbor dispersion metric is highly sensitive to the quality of word vectors, which are in turn dependent on corpus size. Given the sparsity of our older corpora, it is likely that the noise in the vectors produced an unreliable proxy, masking any underlying relationship. This highlights a key challenge in applying such methods in low-resource settings.

5.3 Limitations and Future Work

While our pipeline provides a practical baseline for diachronic analysis, we acknowledge several limitations that offer avenues for future work.

The primary limitation is the corpus itself. The older eras are smaller and affected by OCR noise, despite our cleaning efforts. This data imbalance can affect the quality of the word embeddings and the stability of the statistical analyses. Building larger, cleaner, and more balanced historical corpora for Bengali is the most critical next step for the field.

Second, our use of static Word2Vec models suffers from sense conflation. A single vector cannot distinguish between multiple meanings of a word. This is particularly relevant for words like কুল, where an old meaning decays as a new one rises.

Future work should focus on overcoming these limitations. The development of a manually annotated "gold standard" benchmark of semantic change in Bengali would enable more rigorous evaluation of computational models. Furthermore, applying contextualized language models (e.g., BERT) trained on diachronic data could help disentangle different word senses and provide an even more fine-grained view of semantic change.

6 Conclusion

This paper presented the first large-scale computational study of lexical semantic change in Bengali (1950-2025). We constructed a diachronic corpus and used a hybrid n-gram and embedding pipeline to map semantic evolution, identifying both the shift in existing words and the emergence of new ones.

Our results revealed clear and interpretable patterns of change. The case studies of words like ডিজিটাল (digital), কুল (cool), and স্বাধীনতা (independence) demonstrate how the Bengali lexicon has been shaped by the forces of technological innovation, cultural globalization, and national history. Furthermore, our analysis confirmed that Bengali adheres to the universal linguistic law of frequency, where more frequent words exhibit greater semantic stability. While we did not find support for the law of polysemy, we attribute this to methodological limitations stemming from data sparsity rather than a definitive linguistic feature of Bengali.

While constrained by challenges such as OCR noise and corpus imbalance in early eras, this study provides a practical, reproducible baseline for future diachronic research in Bengali. The findings offer valuable insights for historical linguistics and computational lexicography, and the pipeline itself serves as a viable model for exploring the evolution of other under-resourced languages.

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